
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead on long-video inputs because the number of visual tokens scales with the frame count, and self-attention cost grows quadratically in that number. Prevailing efficiency methods reduce this token budget by pruning or merging tokens according to per-token importance or pairwise similarity. Two limitations follow from this design: such criteria model only informational commonality and therefore discard a complementary signal, namely temporal change and transition points, and they operate on spatial or temporal correlations in isolation rather than modeling the two jointly. We formalize a dual criterion to close this gap: similarity is the signal for redundancy, whereas difference is the signal for salient events. Instantiating this criterion, we introduce ST-SimDiff, a training-free token-selection framework. The procedure first casts the full set of visual tokens as nodes of a spatio-temporal graph, providing a single representation over their spatial and temporal associations. It then applies two selection operators in parallel: (1) a similarity-based operator that runs community detection over the graph and retains high-centrality representative tokens, thereby compressing static content; and (2) a temporal-difference-based operator that localizes abrupt inter-frame feature changes and preserves the tokens marking those dynamic transitions. The union of the two token sets encodes both static and dynamic content under a tight token budget. Across long-video benchmarks, the method attains higher accuracy than state-of-the-art baselines while markedly lowering inference-time computation. Our code will be released upon acceptance.

1 INTRODUCTION

Progress in Large Language Models (LLMs) has directly driven the development of Large Vision-Language Models (LVLMs), which now achieve strong performance on image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). In the video setting, a typical LVLM discretizes an input video into a sequence of sampled frames and encodes each frame into hundreds to thousands of visual tokens that are then consumed by the LLM (Bai et al., 2025; Li et al., 2024). This encoding pipeline is effective, but the resulting token count grows multiplicatively with video duration, frame rate, and spatial resolution. Because the attention cost is quadratic in the token count, the growth translates into prohibitive compute and memory requirements, which constrain the deployment of LVLMs in regimes such as long-video analysis and real-time interaction (Fu et al., 2024b).

A body of work addresses this efficiency bottleneck, and the proposed methods fall into two families defined by their selection criterion. The first family is importance-based pruning. FastV (Chen et al., 2024) observes that the attention mass assigned to visual tokens becomes highly concentrated in the deeper transformer layers, so tokens receiving low attention scores at those layers can be discarded with limited impact on the output, reducing the effective sequence length. The second family is similarity-based merging or selection, which exploits the empirical observation that visual tokens are highly correlated—either across temporally adjacent frames or within a single frame. FrameFusion (Fu et al., 2024b) merges mutually similar tokens from neighboring frames to suppress temporal redundancy, whereas VisionZip (Yang et al., 2025) selects dominant tokens directly at the output of the vision encoder.

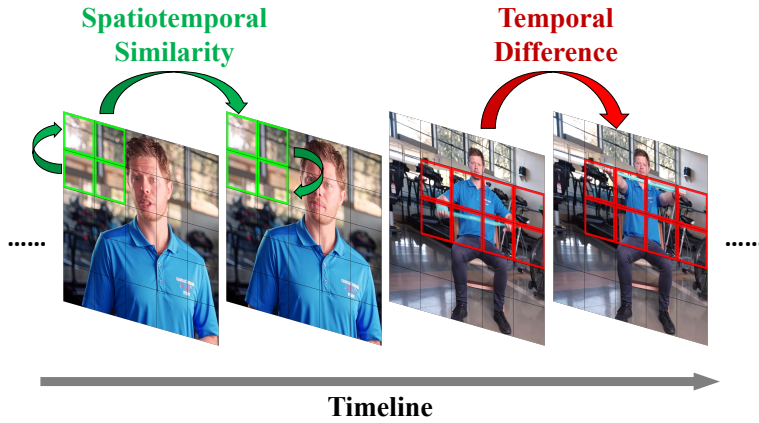


Figure 1: The core motivation of ST-SimDiff. Efficient video understanding requires two operations to be handled concurrently. Spatiotemporal Similarity (left) provides the signal for compression: redundancy is removed both spatially, among tokens within a frame, and temporally, among co-located tokens across adjacent frames. Temporal Difference (right) provides the complementary signal for preservation: abrupt inter-frame changes are detected to retain the key actions or events that define the plot.

Despite their effectiveness at reducing the token budget, existing methods exhibit two structural limitations. First, they treat the spatial and temporal axes separately: pruning is driven either by spatial correlations among tokens within a single frame or by temporal correlations among co-located tokens across frames. Absent a joint model of spatio-temporal correlation, these methods cannot represent the coupled dependencies that characterize complex dynamic events. Second, and more fundamentally, both families optimize a single class of signal—informational commonality, expressed as similarity or importance—and therefore have no mechanism for detecting change. Because a video’s narrative is largely driven by transition events, a criterion that rewards only similarity tends to average over abrupt actions, discarding precisely the tokens on which correct interpretation depends.

Motivated by this analysis, we adopt a dual selection criterion: similarity is the signal that identifies redundancy, and difference is the signal that identifies salient moments. Under this criterion, an ideal token-compression procedure must satisfy two objectives jointly—represent the stable content of a video with as few tokens as possible, and retain the tokens that mark its key changes with high precision. We instantiate the criterion as **ST-SimDiff**, a dual token-selection framework defined over a spatio-temporal graph, comprising two operators. The first is **similarity-based representative token selection**: all visual tokens are treated as graph nodes, and edges are weighted by feature similarity along the spatial and temporal axes; a graph community-detection algorithm partitions the graph into densely connected clusters that correspond to stable, persistent visual elements, and from each cluster we retain a small number of high-centrality tokens as representatives. The second is **difference-based event token selection**: we examine the edges aligned with the temporal axis, and when the similarity between co-located tokens in adjacent frames drops sharply we designate the transition a turning point and mark the associated tokens as critical event tokens to be retained. The two token sets are then merged and passed to the LLM. Our contributions can be summarized as follows:

- We formulate token-level similarity and difference as two co-equal selection signals for VLM efficiency, and argue that both must be modeled rather than similarity alone.
- We define a spatio-temporal graph that unifies the spatial and temporal relationships among video tokens in a single representation, and derive a dual token-selection procedure over it that jointly extracts representative tokens and pivotal event tokens.
- We evaluate the method extensively on video-understanding benchmarks; the results show that ST-SimDiff substantially reduces the visual token budget while maintaining or improving downstream accuracy.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid progress of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has driven parallel advances in multimodal understanding and given rise to a range of Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). Architecturally, an LVLM encodes images or video frames into visual tokens with a pre-trained visual encoder and forwards those tokens to an LLM backbone. The encoder and the backbone are bridged by an alignment module, most commonly an MLP projector (Liu et al., 2024a;b) or a Q-Former (Li et al., 2023; Dai et al., 2023). This composition yields strong performance across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in the video domain it extends to longer, higher-resolution, and structurally more complex inputs (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This capability, however, comes at a substantial computational cost, most acutely in video understanding. Because a video is represented as a continuous sequence of frames, the induced visual-token count far exceeds that of a single image and scales with the number of sampled frames, which stresses both the training and inference efficiency of LVLMs. Reducing the computational overhead of processing large video token sequences therefore remains an open problem in LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Given the overhead of video processing, visual token compression has become a central mechanism for improving LVLM efficiency. Existing approaches can be organized by the quantity they use to score tokens: token importance and token similarity.

Early methods framed the problem as importance-based pruning, which estimates each token’s contribution to the output and removes the low-scoring tokens. Importance is typically quantified from attention scores or related feature statistics, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). A subsequent line of work (Shang et al., 2024; Ye et al., 2025) augments this criterion with token similarity to suppress redundancy more effectively while retaining critical information, on the observation that high-importance tokens can still be mutually similar and hence informationally redundant. Concretely, FrameFusion (Fu et al., 2024b) composes similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) selects dominant tokens and folds contextual information into them; and VisPruner (Zhang et al., 2025b) exploits visual cues to eliminate redundant tokens.

Notwithstanding this progress on token redundancy, prior work leaves the global, coupled spatio-temporal similarity structure among tokens largely unmodeled. In video, redundancy arises jointly along the spatial and temporal axes, yet similarity-driven methods such as FrameFusion restrict attention to temporal similarity between adjacent frames and do not exploit the full spatio-temporal correlation structure. This restriction limits the redundancy that can be removed and, in turn, degrades performance under aggressive token-reduction budgets.

3 PROBLEM DEFINITION

Consider a video V paired with a text query Q . A standard Large Vision-Language Model (LVLM) encodes V into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. Processing this sequence is costly because the self-attention operator has complexity $O(N^2)$ in the token count N . We therefore formulate token reduction as the design of a selection function $f(\cdot)$ that maps the full sequence T_{full} and a target compression ratio r to a substantially smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The optimization objective is to maximize the LVLM’s downstream task performance evaluated on T_{sub} , subject to the budget constraint $|T_{\text{sub}}| \approx r \cdot N$. The central difficulty is thus to specify $f(\cdot)$ so that T_{sub} retains the information most predictive of the downstream task under this constraint.

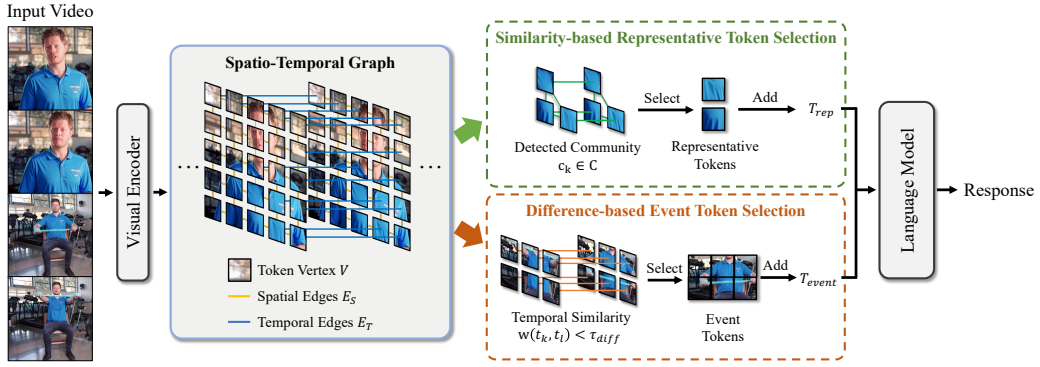


Figure 2: Overview of the ST-SimDiff framework. The pipeline comprises three stages: Spatio-Temporal Graph Construction, which encodes token relations as a weighted graph; Similarity-based Representative Token Selection, which compresses redundant clusters; and Difference-based Event Token Selection, which preserves tokens at high-difference temporal transitions.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As shown in Figure 2, ST-SimDiff operates over a spatio-temporal graph constructed from the video’s visual tokens and applies two selection operators in parallel: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The SRTS operator compresses the stable, redundant content of the video. It applies community detection to the graph, partitioning highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and from each cluster it retains the few tokens of highest centrality to form the representative set T_{rep} . The DETS operator recovers the dynamic content that SRTS is designed to compress away. It examines the temporal edges of the graph and flags positions at which the similarity between co-located tokens in adjacent frames falls sharply; these high-difference positions denote turning points, and the tokens bracketing each transition are retained as critical event tokens, forming the set T_{trans} . Combining the two operators, ST-SimDiff returns the token subset $T_{sub} = T_{rep} \cup T_{trans}$, which is compact yet information-preserving: it encodes stable content through representative tokens while retaining the tokens that mark key events.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder produces a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ is associated with a feature vector $\mathbf{x}_k$ and occupies a fixed spatio-temporal location, which we index by its frame position $T(t_k)$, its spatial row $H(t_k)$, and its spatial column $W(t_k)$.

We encode the pairwise relationships among these tokens as a spatio-temporal graph $G = (V, E)$ whose vertex set V is in one-to-one correspondence with the tokens in T . The edge set decomposes as $E = E_S \cup E_T$, isolating the spatial and temporal relationships. The spatial edge set E_S links tokens that are spatially adjacent within the same frame, i.e., neighbors at unit ℓ_1 distance in the row-column grid:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edge set E_T links tokens that share a spatial position across consecutive frames, thereby encoding temporal continuity at each grid location:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each edge $w(v_i, v_j) \in E$ is weighted by the cosine similarity between the feature vectors of its endpoint tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

Because each vertex is incident to only a constant number of spatial and temporal neighbors, G is sparse ($|E| = O(N)$); it compactly encodes local spatial adjacency and per-position temporal continuity, and serves as the shared representation on which both selection operators are defined.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content exhibits pronounced spatio-temporal redundancy. In a static scene that persists for several seconds, co-located background tokens remain highly similar across frames; for an object translating across the field of view, its tokens change spatial position yet retain strong semantic correlation. Methods that treat the spatial and temporal axes in isolation capture only one of these regularities. By contrast, the spatio-temporal graph G represents both jointly: tokens that are semantically related aggregate into dense communities under the similarity-weighted edges, independent of the specific frame or grid position they occupy. Selecting a small set of representatives from each community therefore constitutes an efficient compression scheme that removes redundancy while preserving the community’s core semantics.

To isolate strongly correlated clusters, we first sparsify G by thresholding its edge weights at a similarity level τ_{sim} , retaining only edges whose weight exceeds the threshold. This yields the pruned graph $G' = (V, E')$ with $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. We then run a graph community-detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to obtain a partition into token clusters $C = \{c_1, c_2, \dots, c_m\}$. Within each community $c_k \in C$, tokens are ranked by centrality, where the centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is its mean similarity to the remaining members of the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We set the intra-community retention rate equal to the global compression ratio r and, from each community c_k , keep the $\lceil |c_k| \cdot r \rceil$ tokens of highest centrality. Applying a common per-community rate handles all cluster sizes uniformly, including singleton communities of unique tokens, for which the ceiling guarantees that the sole member is retained. Formally, the representative token set T_{rep} is

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where TopK returns the k highest-scoring elements of its argument set under the centrality score S_c . In effect, this operator compresses the static, persistent content of the video by retaining only the most central token of each semantic cluster as its surrogate.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

Where similarity characterizes the stationary component of a video, difference characterizes its events. A criterion tuned to similarity alone models “what is” but is insensitive to “what happened,” yet a video’s plot is carried by events, and an event is by definition a change—the appearance of a new object, the onset of an action, or a scene cut. Under our representation, such a change surfaces as a sharp drop in similarity between temporally adjacent tokens. Localizing these high-difference turning points is therefore necessary for modeling the dynamic content and for answering “when” and “why” queries.

Difference along the temporal axis is a strong indicator of a key event: the entry of a new object, a scene transition, or the start or end of an action induces a large feature shift at the corresponding position across adjacent frames. Accordingly, the DETS operator examines only the temporal edges E_T of the spatio-temporal graph. For a temporal edge $(v_k, v_l) \in E_T$ connecting temporally adjacent tokens t_k and t_l , we apply a difference threshold τ (e.g., the 95th percentile of the temporal-edge difference scores). When the edge weight falls below the threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, the low

similarity signals a transition and the later token t_l is retained as a critical event token. Formally, the event token set T_{event} is

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This operator thus preserves exactly the tokens located at points of significant temporal change. By retaining these transitions explicitly, the compression stage cannot discard the dynamic content that defines the video’s narrative.

4.5 OVERALL REDUCTION PROCESS

The framework computes the representative set T_{rep} and the event set T_{event} in parallel and forms their union $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$ as the initial candidate set. To match the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ exactly, we append a final pruning step: whenever $|T_{\text{candidate}}| > N_{\text{target}}$, we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ lowest-importance tokens, where, following (Chen et al., 2024), token importance is measured by the attention score in the shallow layers of the LLM. This step lets the method meet an arbitrary token budget exactly while preferentially discarding the least informative candidates.

The overall procedure therefore composes two complementary mechanisms: graph-based token selection and attention-based budget pruning. The former exploits the intrinsic structure of the video content—its similarity and difference—to guarantee that core information is retained, while the latter provides an adjustable control that enforces the exact budget. The combination makes the compression both content-aware and budget-adaptive, yielding a controlled trade-off between efficiency and accuracy.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

We now bound the computational cost of ST-SimDiff, decomposing it into the three stages of the pipeline: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Graph construction computes, for each of the N tokens, the cosine similarity to its spatial and temporal neighbors. Since every token has a constant number of neighbors, the edge weights are obtained in a single pass over the token set, at cost $O(Nd)$, where d is the token feature dimension.

SRTS then partitions the graph into clusters. As specified in our implementation (Section 5.2), we use a connected-components algorithm with near-linear cost $O(N + |E|)$; because $|E| \leq 3N$, this reduces to $O(N)$. The subsequent intra-community filtering is the only step at risk of superlinear cost: computing all-pairs similarity within every community c_k would cost $O(|C| \cdot |c_k|^2 \cdot d)$, which is quadratic in the community size. We bound this term by capping the community size—whenever a community exceeds a preset threshold (e.g., $|c_k| > \sqrt{N}$) it is partitioned—so the filtering step remains subquadratic and does not dominate. Louvain- and Leiden-style algorithms yield richer community structure but at higher cost, approaching $O(N \log N)$; we therefore adopt connected components to keep the selection overhead minimal.

DETS performs a single pass over the temporal edges, thresholding each edge weight to flag transitions, at linear cost $O(Nd)$.

Summing the three stages, the total cost of ST-SimDiff is dominated by linear-time operations and is $O(Nd)$. This is asymptotically smaller than the $O(N^2d)$ self-attention cost of the LLM backbone, so the token-reduction stage adds only a negligible overhead relative to the model’s dominant inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines We adopt LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and benchmark ST-SimDiff against representative state-of-the-art video token-compression methods spanning two strategy classes: **importance-based reduction**, namely FastV (Chen et al.,

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Accuracy on long-form video understanding benchmarks with LLaVA-Video-7B at 64 input frames (%). For each column, the best score across methods is set in **bold**.

2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction** that couples importance with similarity, namely VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To bracket the achievable range, we further report the uncompressed model (Vanilla) as an upper bound and Random Sampling of an equal token count as a lower bound.

Benchmarks We evaluate long-video understanding on three challenging datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a broad-coverage benchmark for general video understanding, spanning diverse video types and stratified into three duration categories—Short, Medium, and Long. LongVideoBench targets fine-grained retrieval and reasoning over long videos via a “referring reasoning” task, with videos drawn from four duration bins: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering benchmark over long-horizon egocentric video that probes a model’s ability to infer character intent and the causal structure of actions.

5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs (48 GB VRAM) under Ubuntu 22.04, and inference is evaluated with the *lmms-eval* (Zhang et al., 2024a) library. LLaVA-Video uses its default input length of 64 frames; to hold the input budget fixed across models for a controlled efficiency comparison, we set NVILA to 64 input frames as well. The selection hyperparameters are fixed to a similarity threshold $\tau_{\text{sim}} = 0.8$ and a difference threshold $\tau_{\text{diff}} = 0.2$, and we report results at two token-compression ratios, $r = 30\%$ and $r = 50\%$. For community detection we instantiate the generic operator of Section 5 with connected-component finding, which we select on the observed trade-off between runtime and downstream accuracy.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We evaluate ST-SimDiff against a set of mainstream efficient video-VLM compression methods on the three long-form video understanding benchmarks: VideoMME, LongVideoBench, and

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Accuracy on long-form video understanding benchmarks with NVILA-Video-8B at 64 input frames (%). For each column, the best score across methods is set in **bold**.

EgoSchema. To assess generalization and robustness, we report per-benchmark accuracy on two base models, LLaVA-Video-7B and NVILA-8B, at token retain ratios of 30% and 50%. Results appear in Table 1 and Table 2.

(1) ST-SimDiff attains the best score in every tested configuration. Notably, at the 50% retain ratio the overall accuracy on both base models not only exceeds that of competing compression methods but, on several benchmarks, matches or surpasses the uncompressed model that retains 100% of the tokens. This indicates that the selection criterion discards tokens that are net-redundant for the downstream task, and motivates the following analysis of where the baselines fall short.

(2) Among the importance-based baselines (FastV, FasterVLM), FasterVLM is generally the stronger. Their shared weakness is inefficient handling of temporal redundancy: by scoring tokens on importance alone, they retain high-importance but temporally repeated tokens, which lowers the effective compression rate. The hybrid baselines (PruMerge, VisionZip, FrameFusion) are more competitive than the importance-only methods, with FrameFusion the strongest overall. Their objective, however, remains the identification and retention of representative “commonality” information, so they share a structural blind spot: tokens carrying narrative information localized at “turning points” and “abrupt events” receive no explicit protection and may be pruned.

(3) In sum, the baselines encode an increasingly refined notion of “redundancy,” yet leave two issues open: they do not jointly analyze spatio-temporal similarity across the two axes, and so cannot represent the coupled redundancy structure; and they omit an explicit “difference” signal, and so do not actively preserve the changes that define plot development. ST-SimDiff addresses both by modeling the spatio-temporal relations in a single graph and by adding a dedicated difference-detection operator, which together account for its accuracy gains.

5.4 ABLATION STUDY

To isolate the contribution of each operator in ST-SimDiff, we run a controlled ablation that starts from a Baseline using only importance-based pruning and additively enables the similarity operator

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component-wise ablation of ST-SimDiff on LLaVA-Video-7B (%), reporting the incremental contribution of the similarity and difference operators at $r = 30\%$ and $r = 50\%$.

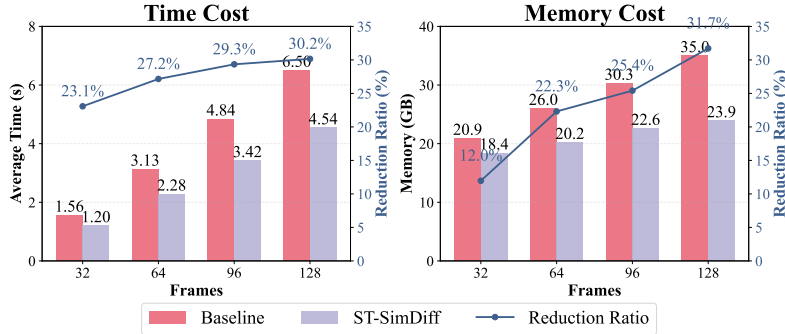


Figure 3: Computational cost of ST-SimDiff versus the uncompressed LLaVA-Video baseline, measured as average per-inference latency and peak GPU memory under a 30% token budget across input lengths from 32 to 128 frames.

(+Sim) and then the difference operator (++Diff), quantifying the marginal effect of each. Both retain ratios, 30% and 50%, are evaluated; results are in Table 5.

(1) We first isolate the +Sim operator. Adding similarity modeling to the Baseline yields a consistent accuracy gain across benchmarks. We further decompose the similarity criterion into three variants—spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.)—and observe that the joint variant is uniformly the strongest, confirming that modeling the two axes together removes redundancy more effectively than either axis alone.

(2) Building on the joint-similarity operator, we enable the difference operator (++Diff) to obtain the complete framework. This addition produces the largest single increment, and the increment is most pronounced at the aggressive $r = 30\%$ ratio. The two operators are thus complementary: +Sim compresses redundancy aggressively, while ++Diff acts as a safeguard that recovers the unique event tokens carrying the video’s narrative. At $r = 50\%$ the increment is smaller, because the looser budget lets +Sim already retain many event tokens and overall accuracy is near its upper bound. This pattern is consistent with our central hypothesis—the framework’s advantage comes from balancing similarity-based redundancy compression against difference-based event preservation, and that balance matters most under tight token budgets.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency of ST-SimDiff, we measure per-inference latency and peak GPU memory under a 30% token budget and compare against the uncompressed baseline (Baseline LLaVA-Video). Figure 3 reports both metrics as a function of input length, sweeping from 32 to 128 frames.

Inference Time ST-SimDiff yields a consistent latency reduction that widens with input length. The baseline’s average inference time rises steeply from 1.56 s at 32 frames to 6.50 s at 128 frames, whereas ST-SimDiff limits the 128-frame cost to 4.54 s. The relative speedup therefore grows with sequence length, from 23.0% at 32 frames to 30.2% at 128 frames—consistent with the quadratic dependence of attention on token count.

Memory Cost The memory trend is analogous. The baseline’s peak GPU memory scales roughly linearly with input length, from 20.9 GB at 32 frames to 35.0 GB at 128 frames. By reducing the number of tokens forwarded to the LLM, ST-SimDiff holds the 128-frame peak to 23.9 GB, a 31.7% reduction relative to the baseline. This lowered memory footprint admits longer inputs on memory-constrained hardware.

In summary, ST-SimDiff attains leading accuracy while simultaneously reducing both latency and peak memory, and the efficiency margin increases with input length.

6 CONCLUSION

This paper addresses the computational inefficiency of Multimodal Large Language Models on long-video inputs, which stems jointly from the large visual-token count and from the fact that existing token-reduction criteria model commonality while ignoring content change. We introduce ST-SimDiff, a training-free framework that represents the visual tokens as a spatio-temporal graph and applies two complementary selection operators over it: a similarity-based operator that identifies redundancy through community detection, and a temporal-difference-based operator that preserves tokens at key transition points. Empirically, on VideoMME and LongVideoBench, ST-SimDiff reduces inference latency and peak memory while consistently exceeding state-of-the-art accuracy. More broadly, the work formalizes token selection as the joint optimization of content representativeness and transition preservation, yielding a more information-efficient representation of video for downstream reasoning.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

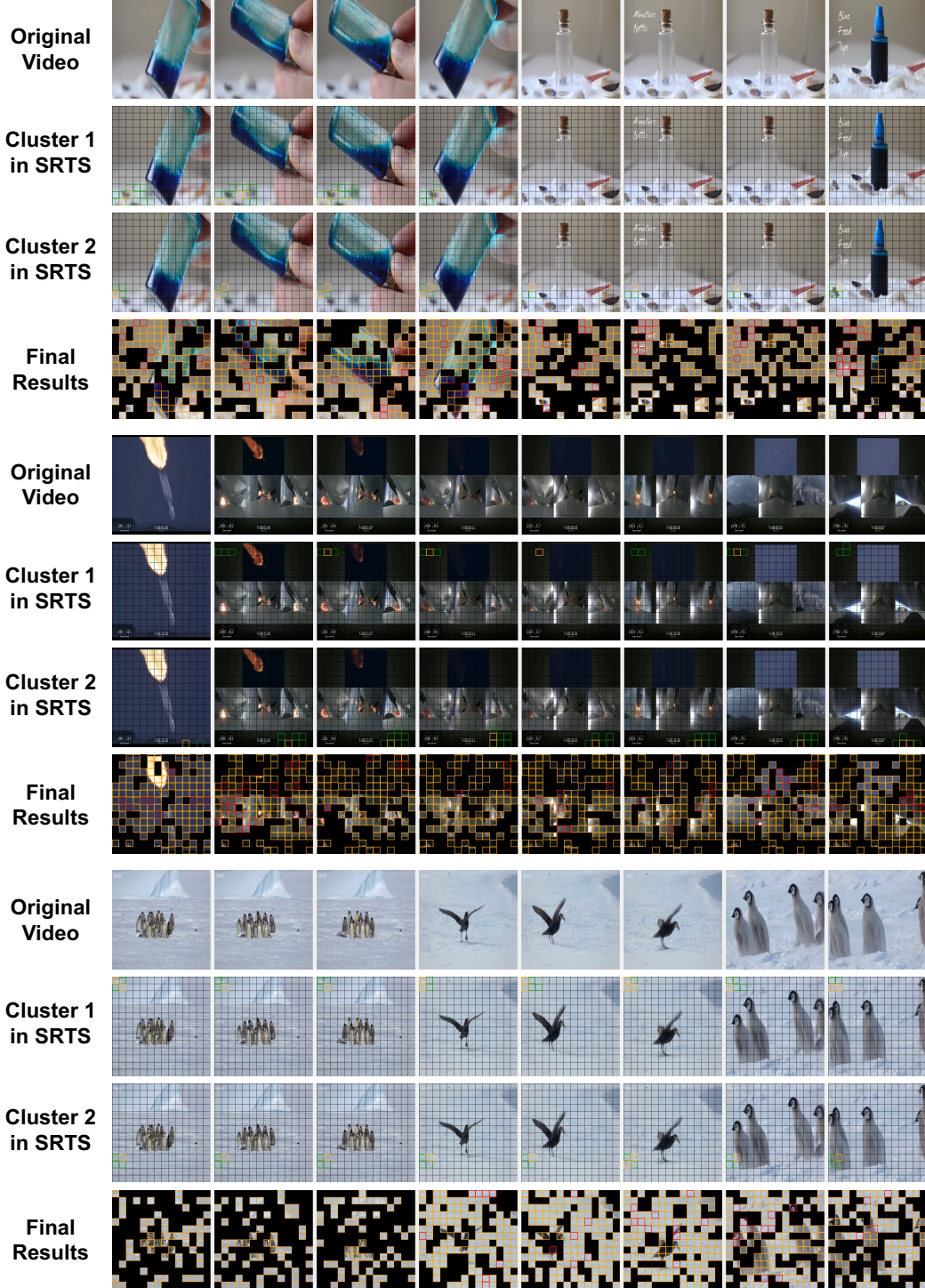


Figure 4: Visualization of the ST-SimDiff selection process. The token selection is decomposed into the communities returned by SRTS (rendered as grids in Cluster 1 and Cluster 2) and the event-driven tokens flagged by DETS. The final panel makes the complementarity explicit: sparse representative tokens (yellow) encode stable content, while dense event tokens (red) capture dynamic actions.

To make the behavior of ST-SimDiff concrete, Figure 4 visualizes the intermediate and final token-selection results on a representative sequence involving a dynamic object-manipulation task. Consider first the Similarity-based Representative Token Selection (SRTS) operator: the rows labeled “Cluster 1” and “Cluster 2” expose the output of the graph community-detection step, which groups spatially and temporally redundant regions—the static background and stationary objects—into coherent communities. The visualization confirms that SRTS localizes stable visual elements and compresses each community to a few central representative tokens, shown as the sparse yellow boxes in the final panel, so that large amounts of background redundancy are handled at low token cost.

The Difference-based Event Token Selection (DETS) operator is active at the dynamic moments. In the “Final Results” row, tokens flagged by high temporal difference are marked with dense red bounding boxes. These boxes coincide with the moving hand and the shifting blue liquid, confirming that the operator localizes the turning points at which the temporal difference exceeds the threshold (τ_{diff}). Consequently, fine-grained action detail and rapid visual change are retained rather than averaged away by the compression stage.

The overlay in the final results makes the interaction of the two operators explicit: sparse yellow tokens cover the stable background, while dense red tokens concentrate on the region of rapid motion. This spatial separation is direct evidence for our design principle. By sampling densely where change is high, ST-SimDiff avoids the blind spot of similarity-only methods; by pruning the joint spatio-temporal redundancy, it removes the repetition that importance-based methods retain. The result is a controlled trade-off between computational cost and the preservation of task-critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Accuracy on long-form video understanding benchmarks with Qwen2.5-VL-7B at 64 and 128 input frames (%).

To test whether the gains of ST-SimDiff transfer across base architectures, we run additional experiments on the recently released Qwen2.5-VL model, evaluating on VideoMME and LongVideoBench. This isolates the effect of the base model and verifies that the method’s efficacy is not tied to a single architecture.

The results in Table 4 show that ST-SimDiff remains strong on Qwen2.5-VL. At a 30% token retention ratio it reaches 62.4 on VideoMME and 56.8 on LongVideoBench; raising the budget to a 50% retention ratio increases these to 62.9 and 58.4, respectively. Notably, at the 50% ratio the VideoMME accuracy equals the all-token upper bound and the LongVideoBench accuracy is within a narrow margin of its full-token counterpart, corroborating the cross-architecture generalization of the framework.

C ABLATION STUDY RESULTS ON NVILA

To verify that the contribution of each operator transfers across base models, we repeat the component ablation on the NVILA-Video-8B model (Liu et al., 2024c). The protocol matches the ablation

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Component-wise ablation of ST-SimDiff on the NVILA-Video-8B model (%), reporting the incremental contribution of the similarity and difference operators at $r = 30\%$ and $r = 50\%$.

in the main paper: starting from a Baseline that applies only importance-based pruning, we additively enable the similarity operator (+ Sim) and then the difference operator (++ Diff).

Table 5 shows that the trends observed on LLaVA-Video-7B recur on NVILA-Video-8B, confirming the design across models. Enabling the +Sim operator over the Baseline yields a steady accuracy gain at both the 30% and 50% retain ratios. Within +Sim we compare three variants—spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.)—and the joint variant is the strongest: at $r = 50\%$ it attains 61.4% on VideoMME, exceeding both single-axis variants. This restates the point that jointly modeling the spatio-temporal associations is the more effective way to identify redundant tokens. Adding the difference operator (++ Diff) on top of the joint-similarity operator completes the framework and produces the largest single increment in the table. The pattern substantiates the generalization of the two core operators—similarity and difference selection—and supports the central claim that redundancy and change must be handled jointly.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

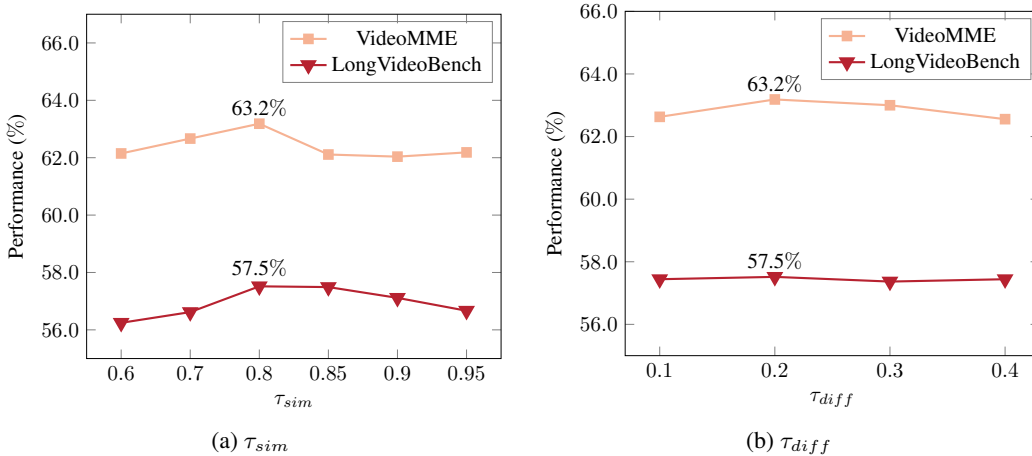


Figure 5: Sensitivity of accuracy to the thresholds τ_{diff} and τ_{sim} on VideoMME and LongVideoBench (LLaVA-Video).

The similarity threshold τ_{sim} and the difference threshold τ_{diff} are the two hyperparameters that govern, respectively, the selection of representative and transitional tokens. We sweep each threshold and report accuracy in Figure 6. Both curves are unimodal—accuracy rises and then falls—and remain nearly flat over a broad interval, indicating low sensitivity. The mechanism behind each extreme is interpretable: for τ_{sim} , too low a value under-sparsifies the graph and yields imprecise communities, whereas too high a value fragments coherent semantic clusters; for τ_{diff} , too low a value makes the detector over-sensitive and admits noise, whereas too high a value suppresses genuine events. Accuracy is maximized at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, which we fix for all remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

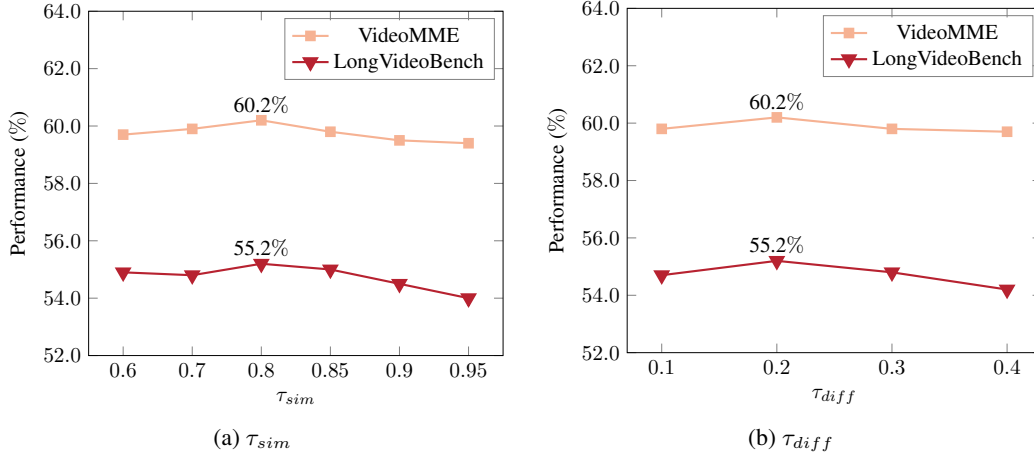


Figure 6: Sensitivity of accuracy to the thresholds τ_{diff} and τ_{sim} on the NVILA-Video-8B model.

To characterize the sensitivity of the two key thresholds on a second base model, we repeat the sweep on NVILA-Video-8B. Figure 6 reports accuracy as a function of the similarity threshold τ_{sim} and the difference threshold τ_{diff} on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . Sweeping τ_{sim} over $[0.6, 0.95]$, we find that accuracy varies little with the threshold. Both benchmarks peak at $\tau_{sim} = 0.8$ —60.2% on VideoMME and 55.2% on LongVideoBench—and the degradation at the interval endpoints is small, so the method is stable across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . Sweeping τ_{diff} over $[0.1, 0.4]$, we observe the same robustness. Accuracy is maximized near $\tau_{diff} = 0.2$ on both benchmarks, at 60.2% on VideoMME and 55.2% on LongVideoBench, indicating that a threshold near 0.2 best trades off event capture against token count.

Taken together, these sweeps establish that the method is insensitive to its two thresholds within the tested ranges and justify the choice $\tau_{sim} = 0.8$, $\tau_{diff} = 0.2$ used in the main experiments, which is jointly optimal across benchmarks and base models.